

A COUNTEREXAMPLE TO EXACT QUADRATIC CONTAINMENT IN GREEN'S PROBLEM 47

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ABSTRACT. We give a counterexample to the exact-containment formulation in the comments to Green's Problem 47. The set

$$A = \{2\} \cup \{q^2 : q \text{ is prime and } q \equiv 1 \pmod{8}\}$$

occupies at most $(p+1)/2$ residue classes modulo every odd prime p , but $|A \cap [0, X]|$ is not $O(\sqrt{X}/(\log X)^{100})$ and A is not contained in $P(\mathbb{Z})$ for any quadratic polynomial $P \in \mathbb{Q}[t]$. The obstruction to quadratic containment is a two-moment identity for quadratic images over finite fields. The counting argument uses divergence of a prime series and a dyadic decomposition, without a prime number theorem. We give a complete Lean formalization and record local checks against the precise counterexample target used by Conjectures.io. Removing the single element 2 leaves a subset of the squares; thus this construction does not refute a formulation allowing finitely many exceptional elements, including Green and Harper's Conjecture 1.7.

1. INTRODUCTION

For $A \subseteq \mathbb{N}$ and a prime p , write

$$A \bmod p = \{a \bmod p : a \in A\} \subseteq \mathbb{F}_p, \quad A(X) = |\{a \in A : a \leq X\}|.$$

Here $\mathbb{N} = \{0, 1, 2, \dots\}$ and logarithms are natural. All elements of the set constructed below are positive, so the choice between $[0, X]$ and $[1, X]$ does not affect its counting function.

The comments to Problem 47 in Green's problem list [1, p. 23] ask whether the condition

$$|A \bmod p| \leq \frac{p+1}{2} \quad \text{for every sufficiently large prime } p \tag{1.1}$$

forces either

$$A(X) = O\left(\frac{\sqrt{X}}{(\log X)^{100}}\right) \quad (X \rightarrow \infty) \tag{1.2}$$

or $A \subseteq P(\mathbb{Z})$ for some polynomial $P \in \mathbb{Q}[t]$ of degree exactly two. The corresponding formal statement is recorded in the Formal Conjectures library [3] and is used by Conjectures.io [4]. The word “every” in the containment conclusion is decisive.

Theorem 1.1. *Let*

$$\mathcal{P}_1 = \{q \in \mathbb{N} : q \text{ is prime and } q \equiv 1 \pmod{8}\}, \quad A = \{2\} \cup \{q^2 : q \in \mathcal{P}_1\}. \tag{1.3}$$

Then the following assertions hold.

- (i) *For every odd prime p , $|A \bmod p| \leq (p+1)/2$.*
- (ii) *$A(X)$ does not satisfy (1.2).*
- (iii) *For every $P \in \mathbb{Q}[t]$ of degree two, $A \not\subseteq P(\mathbb{Z})$.*

Consequently, the exact-containment implication above is false.

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The first few elements of A are

$$2, 289, 1681, 5329, 7921, 9409, 12769, 18769, 37249, \dots$$

The construction starts with selected prime squares and adds one nonsquare. At a prime where 2 is a quadratic nonresidue, none of the selected primes is zero modulo that prime. The missing zero residue makes room for the added element without violating (1.1). Nevertheless, replacing zero by 2 changes the first two moments in a way that no quadratic image can have. Primes in suitable arithmetic progressions allow this finite-field obstruction to be applied to every rational quadratic.

Green and Harper [2, Conjecture 1.7] allow all but finitely many elements to belong to a rational quadratic image. Their counting alternative also concerns arbitrarily large cutoffs for each logarithmic exponent. Our set already satisfies their structural alternative, because

$$A \setminus \{2\} \subseteq \{z^2 : z \in \mathbb{Z}\}. \quad (1.4)$$

Theorem 1.1 therefore addresses the stated exact-containment implication, not the general inverse large sieve problem. No claim of priority is made for the construction.

2. A MOMENT OBSTRUCTION FOR QUADRATIC IMAGES

Fix an odd prime p , and let

$$Q_p = \{x^2 : x \in \mathbb{F}_p\}, \quad Q_p^* = Q_p \setminus \{0\}.$$

These are sets of values; all sums over them count each value once.

Lemma 2.1. *We have $|Q_p| = (p+1)/2$. If $p > 5$, then, as identities in $\mathbb{F}_p$,*

$$\sum_{r \in Q_p} r = 0, \quad \sum_{r \in Q_p} r^2 = 0. \quad (2.1)$$

Proof. The map $x \mapsto x^2$ has one preimage over zero and two preimages over each nonzero square. Thus $p = 1 + 2(|Q_p| - 1)$.

For $j \in \{2, 4\}$ the polynomial $t^j - 1$ has at most j roots. Since $p - 1 > j$, there is $c \in \mathbb{F}_p^*$ with $c^j \neq 1$. Multiplication by c permutes $\mathbb{F}_p$, so

$$\sum_{x \in \mathbb{F}_p} x^j = \sum_{x \in \mathbb{F}_p} (cx)^j = c^j \sum_{x \in \mathbb{F}_p} x^j.$$

The sum is therefore zero. Counting the fibers of $x \mapsto x^2$ gives

$$2 \sum_{r \in Q_p} r = \sum_{x \in \mathbb{F}_p} x^2 = 0, \quad 2 \sum_{r \in Q_p} r^2 = \sum_{x \in \mathbb{F}_p} x^4 = 0.$$

Division by 2 proves (2.1). □

For a subset $S \subseteq \mathbb{F}_p$, put

$$M_1(S) = \sum_{s \in S} s, \quad M_2(S) = \sum_{s \in S} s^2,$$

with both moments computed in $\mathbb{F}_p$.

Lemma 2.2 (Quadratic image identity). *Let $p > 5$, and let $f(t) = at^2 + bt + c \in \mathbb{F}_p[t]$ with $a \neq 0$. For $S = f(\mathbb{F}_p)$ we have*

$$|S| = \frac{p+1}{2}, \quad M_2(S) = 2M_1(S)^2. \quad (2.2)$$

Proof. Completing the square gives

$$f(t) = a \left(t + \frac{b}{2a} \right)^2 + v, \quad v = c - \frac{b^2}{4a},$$

and hence $S = v + aQ_p$. The affine map $r \mapsto v + ar$ is injective, so $|S| = |Q_p| = (p+1)/2$. When regarded as an element of $\mathbb{F}_p$, the integer $(p+1)/2$ is $1/2$. Lemma 2.1 now yields

$$M_1(S) = \sum_{r \in Q_p} (v + ar) = \frac{v}{2}$$

and

$$M_2(S) = \sum_{r \in Q_p} (v + ar)^2 = \frac{v^2}{2} + 2va \sum_{r \in Q_p} r + a^2 \sum_{r \in Q_p} r^2 = \frac{v^2}{2}.$$

It follows that $M_2(S) = 2M_1(S)^2$. □

Corollary 2.3. *Suppose $p > 5$ and 2 is a quadratic nonresidue modulo p . Then*

$$T_p = Q_p^* \cup \{2\}$$

is not contained in the image of any polynomial of degree two over $\mathbb{F}_p$.

Proof. Since $2 \notin Q_p$, the set T_p has $(p+1)/2$ elements. If $T_p \subseteq f(\mathbb{F}_p)$ for a quadratic f , equality follows from Lemma 2.2. Removing zero from Q_p changes neither moment, whereas adjoining 2 gives

$$M_1(T_p) = 2, \quad M_2(T_p) = 4.$$

The quadratic image identity would give $4 = 2 \cdot 2^2 = 8$ in $\mathbb{F}_p$, which is impossible for $p > 5$. □

3. RESIDUE CLASSES OF THE CONSTRUCTION

We use the supplementary law of quadratic reciprocity

$$\left(\frac{2}{p} \right) = (-1)^{(p^2-1)/8} \quad (p \text{ an odd prime}). \quad (3.1)$$

In particular, 2 is a square modulo p exactly when $p \equiv 1$ or $7 \pmod{8}$.

Proposition 3.1. *The set A in (1.3) satisfies $|A \bmod p| \leq (p+1)/2$ for every odd prime p .*

Proof. If $2 \in Q_p$, every element of A reduces to a member of Q_p , and Lemma 2.1 gives the required bound.

Suppose instead that $2 \notin Q_p$. If $q \in \mathcal{P}_1$, then $q \neq p$: otherwise $p \equiv 1 \pmod{8}$, contrary to (3.1). Since q is prime, its residue modulo p is nonzero. Consequently

$$A \bmod p \subseteq Q_p^* \cup \{2\}, \quad |A \bmod p| \leq \frac{p-1}{2} + 1 = \frac{p+1}{2}.$$

□

The following saturation property will rule out containment in a rational quadratic image. We use Dirichlet's theorem in its usual form: a residue class $r \pmod{m}$ contains arbitrarily large primes whenever $\gcd(r, m) = 1$.

Lemma 3.2. *Let p be an odd prime and $u \in \mathbb{F}_p^*$. There is a prime $q \equiv 1 \pmod{8}$ whose residue modulo p is u . In particular,*

$$Q_p^* \cup \{2\} \subseteq A \bmod p. \quad (3.2)$$

If 2 is a quadratic nonresidue modulo p , equality holds in (3.2).

Proof. Because $\gcd(8, p) = 1$, the Chinese remainder theorem supplies a residue class $r \pmod{8p}$ such that

$$r \equiv 1 \pmod{8}, \quad r \equiv u \pmod{p}.$$

The first congruence makes r coprime to 8, and the second makes it coprime to p . Thus $\gcd(r, 8p) = 1$. Dirichlet's theorem supplies a prime q in this class. Then $q \in \mathcal{P}_1$ and $q^2 \pmod{p} = u^2$.

Every nonzero square therefore occurs in $A \pmod{p}$, and 2 occurs because $2 \in A$. When 2 is a nonresidue, the reverse inclusion is the one proved in Proposition 3.1. $\square$

4. EXCLUDING EVERY RATIONAL QUADRATIC

Proposition 4.1. *There is no polynomial $P \in \mathbb{Q}[t]$ of degree two with $A \subseteq P(\mathbb{Z})$.*

Proof. Suppose such a polynomial exists. Clear its coefficient denominators to write

$$P(t) = \frac{at^2 + bt + c}{d}, \quad a, b, c, d \in \mathbb{Z}, \quad a \neq 0, \quad d \neq 0. \quad (4.1)$$

The assumed containment states that for every $n \in A$ there is $z \in \mathbb{Z}$ such that

$$dn = az^2 + bz + c. \quad (4.2)$$

Dirichlet's theorem provides a prime p with

$$p \equiv 3 \pmod{8}, \quad p > \max(5, |a|, |d|).$$

In particular, the residues of a and d modulo p are nonzero. Reducing (4.2) modulo p and dividing by d gives

$$A \pmod{p} \subseteq f(\mathbb{F}_p), \quad f(t) = \frac{a}{d}t^2 + \frac{b}{d}t + \frac{c}{d} \in \mathbb{F}_p[t],$$

where the displayed coefficients are interpreted in $\mathbb{F}_p$. The leading coefficient is nonzero, so f has degree two.

Since $p \equiv 3 \pmod{8}$, equation (3.1) says that 2 is a quadratic nonresidue modulo p . Lemma 3.2 therefore gives $A \pmod{p} = T_p$. We have obtained $T_p \subseteq f(\mathbb{F}_p)$, contrary to Corollary 2.3. $\square$

This argument uses only reduction of an integer polynomial identity. In particular, it does not require choosing a reduction map on all of $\mathbb{Q}$; the finitely many relevant denominators are handled by (4.1).

5. THE COUNTING ALTERNATIVE FAILS

The analytic input needed below is the classical divergence statement

$$\sum_{\substack{q \text{ prime} \\ q \equiv 1 \pmod{8}}} \frac{\log q}{q} = \infty. \quad (5.1)$$

It follows, for example, from the divergence of the reciprocal-prime series in a reduced arithmetic progression. The weighted statement itself is available in the pinned Mathlib library [5], and is the input used in the Lean proof. No prime-counting asymptotic is needed. The formalized analytic background is part of the L -function development described by Loeffler and Stoll [6].

Proposition 5.1. *The counting function of A does not satisfy (1.2).*

Proof. Put

$$N(Y) = |\{q \in \mathcal{P}_1 : q \leq Y\}|.$$

For integer $Y \geq 1$, the map $q \mapsto q^2$ is injective and takes the primes counted by $N(Y)$ into $A \cap [0, Y^2]$. Hence

$$N(Y) \leq A(Y^2). \quad (5.2)$$

Assume for a contradiction that (1.2) holds. Applying it to $X = 2^{2k}$ and using (5.2) gives a constant $C > 0$ such that, for every sufficiently large integer k ,

$$N(2^k) \leq C \frac{2^k}{k^{100}}. \quad (5.3)$$

Here the constant absorbs the factor $(2 \log 2)^{-100}$, since $\log(2^{2k}) = 2k \log 2$.

For $2^k \leq q < 2^{k+1}$, we have

$$\frac{\log q}{q} \leq \frac{(k+1) \log 2}{2^k}.$$

It follows from (5.3), with $k+1$ in place of k , that

$$\begin{aligned} \sum_{\substack{q \in \mathcal{P}_1 \\ 2^k \leq q < 2^{k+1}}} \frac{\log q}{q} &\leq N(2^{k+1}) \frac{(k+1) \log 2}{2^k} \\ &\leq \frac{2C \log 2}{(k+1)^{99}} \end{aligned} \quad (5.4)$$

for all sufficiently large k . The series of upper bounds converges. The omitted initial blocks contain only finitely many primes. Since all summands are nonnegative, summing (5.4) over the dyadic blocks proves convergence of the series in (5.1). This is a contradiction. $\square$

Proof of Theorem 1.1. The three assertions are Proposition 3.1, Proposition 5.1, and Proposition 4.1, respectively. The same set A satisfies the local hypothesis and violates both proposed conclusions. $\square$

Remark 5.2. The dyadic argument works with any fixed real exponent $\beta > 2$ in place of 100: the final majorant is a constant multiple of $(k+1)^{1-\beta}$. We retain the exponent 100 in the main theorem and in the formalized target to match the stated problem exactly. The parameterized extension in this remark is not part of the supplied Lean theorem.

6. LEAN FORMALIZATION AND VERIFICATION

The accompanying source contains the full proof in Lean 4.33.1 with a pinned Mathlib revision. Two forms are supplied. The standalone project imports Mathlib directly and spells out the target proposition. The submission form, `Main.lean`, uses the imports and outer namespace supplied by the official task wrapper. Both prove the negation of the full implication, with the same explicit set (1.3).

6.1. The precise target. Let G denote the proposition

$$\forall A \subseteq \mathbb{N}, \quad (1.1) \implies ((1.2) \vee \exists P \in \mathbb{Q}[t], \deg P = 2 \wedge A \subseteq P(\mathbb{Z})).$$

The catalogued source, with its answer marker resolved as in the platform's trusted environment, has type $\text{True} \leftrightarrow G$. The counterexample target is $\neg(\text{True} \leftrightarrow G)$, which is equivalent to $\neg G$. The submitted declaration is the following; the three named lemmas contain the substantive arguments developed above.

```
theorem target : ¬ (fcTypeOfName% "Green47.green_47") := by
  intro h
  have hA := h.mp trivial
  Green47Counterexample.counterexample
  Green47Counterexample.local_bound
  exact hA.elim Green47Counterexample.not_small
  Green47Counterexample.no_rational_quadratic
```

The macro `fcTypeOfName%` retrieves the source declaration's *type*. It does not use the unfinished source theorem as a proof. The argument introduces the asserted equivalence, obtains its universal conclusion from `True`, specializes to A , and eliminates the two alternatives using the corresponding negative results.

The coercions hidden by the informal notation are explicit in the standalone file. Reduction modulo p lands in $\mathbb{Z} \bmod p$; the counting function is real-valued on natural cutoffs; the polynomial coefficients are

rational; and the input z to the polynomial ranges over the integers. The upper bound on residue classes is natural-number division, which equals $(p + 1)/2$ for odd primes.

6.2. Proof components. The main local declarations correspond to the following steps.

Lean declaration	Mathematical content
<code>quadratic_moment</code>	The two-moment identity of Lemma 2.2.
<code>local_bound</code>	The eventual residue-class bound of Proposition 3.1.
<code>goodPrime_in_residue</code>	The Chinese remainder and Dirichlet argument of Lemma 3.2.
<code>no_rational_quadratic</code>	Denominator clearing and the finite-field obstruction.
<code>primeWeight_not_summable</code>	The divergence input (5.1).
<code>not_small</code>	The dyadic argument of Proposition 5.1.

The divergence input is Mathlib’s theorem

```
ArithmeticFunction.vonMangoldt.  
not_summable_residueClass_prime_div
```

in `Mathlib/NumberTheory/LSeries/PrimesInAP.lean`. Dirichlet’s existence theorem is used for both residue saturation and the choice of a prime avoiding the integer coefficients. The formal proof partitions all natural numbers using `Nat.log 2`; its prime weight is zero outside $\mathcal{P}_1$, so the initial block and the input zero cause no extra assumptions.

6.3. What was checked. Local Lean compilation accepted the complete source, and the axiom audit of `Bounty.target` returned exactly

```
[propext, Classical.choice, Quot.sound]
```

There is no `sorryAx` in the target’s dependencies and no added axiom in the submitted proof. In addition, the official task inspector confirmed that the elaborated type is the intended counterexample type. The canonical source and target type hashes agree with the task manifest. The submitted source also passed the validator’s static policy checks.

These are local compilation, type, and axiom checks. The platform’s separate production verification pipeline, including its Comparator and kernel-replay stage, was not run as part of the recorded local verification. This distinction does not change the mathematical statement proved here, but it specifies the extent of the reported verification.

The source archive includes the standalone project, the exact submission source and wrapper, the task files, the verification reports, and replay instructions. The principal versions are:

Component	Version or revision
Lean	4.33.1
Mathlib	0df444a360eaa60ab8c11dca51a86af692955474
Formal Conjectures	8432eac998110a563e03df65a28c117e97c8c142
Validator	922a912f3777ac11451e2d145356e78de4dbedb2

The Formal Conjectures revision is the platform’s reconstructed source pin; the archive records how it is obtained from the public base and the official patch. It is not necessary to assume that this generated commit is directly fetchable from the upstream repository.

ACKNOWLEDGMENT OF AI ASSISTANCE

OpenAI Codex, an AI assistant, contributed to developing the construction, organizing the argument, producing the Lean formalization, checking the formal artifacts, and drafting this manuscript. This assistance is explicitly acknowledged. The Lean source and the verification records are supplied so that the mathematical and formalization claims can be checked independently.

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